

The factorization $G(\mathfrak{m}) = HU$ of the Monster group $G(\mathfrak{m})$

Abstract

We give a self-contained account of a machine-checked proof that the group $G(\mathfrak{m})$, defined by generators and relations attached to the Monster Lie algebra, is the product HU of its toral subgroup H and its unipotent subgroup U . This is Theorem 3.6 of the source. The formal version replaces the coefficients of the modular function J by an arbitrary function $c: \mathbb{N} \rightarrow \mathbb{N}$ and imposes no hypothesis on it. The proof shows that HU is a subgroup containing the images of all generators, using that H normalizes U (Theorem 3.4 and Lemma 3.5 of the source). All results presented here, including these auxiliary ones, are machine-verified.

Source	Carbone et al., G-Simple (draft), GSimple_Theorem_3_6
Status	proof machine-verified (Lean 4, Mathlib)
Lemmas	6
Text	written by claude-opus-5-5, checked against the Lean source
Run	srv_20261006_173044_GSimple_Theorem_3_6

Contents

1	Setting	1
2	Lemmas	3
2.1	The torus normalizes the unipotent subgroup	3
2.2	The set HU is a subgroup	4
2.3	Elements of H , of U , and generators lie in HU	4
3	Proof of the theorem	4
A	Correspondence with the formal proof	5

1 Setting

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, $\mathbb{C}^\times$ is the group of nonzero complex numbers, and $c: \mathbb{N} \rightarrow \mathbb{N}$ is an arbitrary function. In the source, $c(j)$ is the coefficient of q^j in $J(q)$.

Definition 1.1 (Index set). Let

$$E_c = \{(\ell, j, k) \in \mathbb{N}^3 \mid \ell < j, 1 \leq k \leq c(j)\}.$$

For $\ell, j \in \mathbb{N}$, put

$$c_{\ell j} = (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C}.$$

Definition 1.2 (Generators). Let $\mathcal{X}$ be the set of symbols

$$H_1(s), \quad H_2(s), \quad X_{-1}(u), \quad Y_{-1}(u), \quad X_{\ell,jk}(u), \quad Y_{\ell,jk}(u),$$

where $s \in \mathbb{C}^\times$, $u \in \mathbb{C}$ and $(\ell, j, k) \in E_c$. Let $F(\mathcal{X})$ be the free group on $\mathcal{X}$. In $F(\mathcal{X})$ define, for $s \in \mathbb{C}^\times$ and $(\ell, j, k) \in E_c$,

$$\begin{aligned} \tilde{w}_{-1}(s) &= X_{-1}(s)Y_{-1}(-s^{-1})X_{-1}(s), & \tilde{w}_{-1} &= \tilde{w}_{-1}(1), \\ \tilde{w}_{\ell,jk}(s) &= X_{\ell,jk}(s)Y_{\ell,jk}\left(\frac{-s^{-1}}{c_{\ell j}}\right)X_{\ell,jk}(s), & \tilde{w}_{\ell,jk} &= \tilde{w}_{\ell,jk}(1). \end{aligned}$$

The commutator is $(a, b) = aba^{-1}b^{-1}$.

Definition 1.3 (The group $G_c(\mathfrak{m})$). Let $\mathcal{R} \subseteq F(\mathcal{X})$ consist of the following relators. An equation $a = b$ stands for the relator ab^{-1} , and a commutator (a, b) stands for itself. The relators are taken for all $s, t \in \mathbb{C}^\times$, $u, v \in \mathbb{C}$ and all indices satisfying the stated membership conditions.

$$\begin{aligned} (\text{H:1a}) \quad & H_1(s)H_1(t) = H_1(st), & (\text{H:1b}) \quad & H_2(s)H_2(t) = H_2(st), & (\text{H:2}) \quad & H_1(s)H_2(t) = H_2(t)H_1(s), \\ (\text{Re:1a}) \quad & X_{-1}(u)X_{-1}(v) = X_{-1}(u+v), & (\text{Re:1b}) \quad & Y_{-1}(u)Y_{-1}(v) = Y_{-1}(u+v), \\ (\text{Re:2}) \quad & Y_{-1}(-t)X_{-1}(s)Y_{-1}(t) = X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1}), \\ (\text{Re:3}) \quad & \tilde{w}_{-1}(s)\tilde{w}_{-1} = H_1(-s)H_2(-s^{-1}), \\ (\text{Re:4a}) \quad & \tilde{w}_{-1}X_{-1}(u)\tilde{w}_{-1}^{-1} = Y_{-1}(-u), & (\text{Re:4b}) \quad & \tilde{w}_{-1}Y_{-1}(u)\tilde{w}_{-1}^{-1} = X_{-1}(-u), \\ (\text{Re:5a}) \quad & \tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1} = H_2(s), & (\text{Re:5b}) \quad & \tilde{w}_{-1}H_2(s)\tilde{w}_{-1}^{-1} = H_1(s), \\ (\text{Re:6a}) \quad & H_1(s)X_{-1}(u)H_1(s)^{-1} = X_{-1}(su), & (\text{Re:6b}) \quad & H_2(s)X_{-1}(u)H_2(s)^{-1} = X_{-1}(s^{-1}u), \\ (\text{Re:6c}) \quad & H_1(s)Y_{-1}(u)H_1(s)^{-1} = Y_{-1}(s^{-1}u), & (\text{Re:6d}) \quad & H_2(s)Y_{-1}(u)H_2(s)^{-1} = Y_{-1}(su). \end{aligned}$$

For $(\ell, j, k) \in E_c$, writing $X = X_{\ell,jk}$, $Y = Y_{\ell,jk}$, $\tilde{w} = \tilde{w}_{\ell,jk}$ and $\gamma = c_{\ell j}$:

$$\begin{aligned} (\text{Im:1a}) \quad & X(u)X(v) = X(u+v), & (\text{Im:1b}) \quad & Y(u)Y(v) = Y(u+v), \\ (\text{Im:2}) \quad & Y(-t)X(s)Y(t) = X\left(\frac{-t^{-1}}{\gamma}\right)Y(-\gamma t^2s)X\left(\frac{t^{-1}}{\gamma}\right), \\ (\text{Im:3}) \quad & \tilde{w}\left(s^{(\ell+1)(j-\ell)}\right)\tilde{w} = H_1\left((-s)^{j-\ell}\right)H_2\left((-s)^{\ell+1}\right), \\ (\text{Im:4a}) \quad & \tilde{w}X(u)\tilde{w}^{-1} = Y(-u/\gamma), & (\text{Im:4b}) \quad & \tilde{w}Y(u)\tilde{w}^{-1} = X(-\gamma u), \\ (\text{Im:5a}) \quad & \tilde{w}H_1(s^{j-\ell})\tilde{w}^{-1} = H_2\left((s^{\ell+1})^{-1}\right), & (\text{Im:5b}) \quad & \tilde{w}H_2(s^{\ell+1})\tilde{w}^{-1} = H_1\left((s^{j-\ell})^{-1}\right), \\ (\text{Im:6a}) \quad & H_1(s)X(u)H_1(s)^{-1} = X(s^{\ell+1}u), & (\text{Im:6b}) \quad & H_2(s)X(u)H_2(s)^{-1} = X(s^{j-\ell}u), \\ (\text{Im:6c}) \quad & H_1(s)Y(u)H_1(s)^{-1} = Y\left((s^{\ell+1})^{-1}u\right), & (\text{Im:6d}) \quad & H_2(s)Y(u)H_2(s)^{-1} = Y\left((s^{j-\ell})^{-1}u\right). \end{aligned}$$

The relations among the unipotent generators are:

$$\begin{aligned} (\text{U:1a}) \quad & (X_{-1}(u), X_{j-1,jk}(v)) = 1, & (\text{U:1b}) \quad & (Y_{-1}(u), Y_{j-1,jk}(v)) = 1 \quad \text{for } (j-1, j, k) \in E_c, \\ (\text{U:1c}) \quad & (Y_{-1}(u), X_{0,jk}(v)) = 1, & (\text{U:1d}) \quad & (X_{-1}(u), Y_{0,jk}(v)) = 1 \quad \text{for } (0, j, k) \in E_c, \\ (\text{U:2}) \quad & (X_{\ell,jk}(u), Y_{m,pq}(v)) = 1 \quad \text{for } (\ell, j, k), (m, p, q) \in E_c \text{ with } j \neq p \text{ or } k \neq q \text{ or } |\ell - m| > 1, \\ (\text{U:3a}) \quad & (X_{1,2k}(u), Y_{0,2k}(v)) = X_{-1}(uv), & (\text{U:3b}) \quad & (X_{0,2k}(u), Y_{1,2k}(v)) = Y_{-1}(uv) \quad \text{for } (0, 2, k), (1, 2, k) \in E_c, \\ (\text{U:4a}) \quad & \tilde{w}_{-1}X_{\ell,jk}(u)\tilde{w}_{-1}^{-1} = X_{j-1-\ell,jk}\left((-1)^{j-\ell-1}u\right), \\ (\text{U:4b}) \quad & \tilde{w}_{-1}Y_{\ell,jk}(u)\tilde{w}_{-1}^{-1} = Y_{j-1-\ell,jk}\left((-1)^{j-\ell-1}u\right) \quad \text{for } (\ell, j, k), (j-1-\ell, j, k) \in E_c. \end{aligned}$$

All exponents of s and of -1 are natural numbers. Then

$$G_c(\mathfrak{m}) = F(\mathcal{X})/N_{\mathcal{R}},$$

where $N_{\mathcal{R}}$ is the normal closure of $\mathcal{R}$. We use the same symbols for the images of generators in $G_c(\mathfrak{m})$.

Definition 1.4 (Toral and unipotent subgroups). Let $T_H \subseteq G_c(\mathfrak{m})$ be the set of images of all $H_1(s)$ and $H_2(s)$, $s \in \mathbb{C}^\times$. The *toral subgroup* is $H = \langle T_H \rangle$.

Let $T_U \subseteq G_c(\mathfrak{m})$ be the set of images of all $X_{-1}(u)$, $Y_{-1}(u)$, $X_{\ell,jk}(u)$ and $Y_{\ell,jk}(u)$, for $u \in \mathbb{C}$ and $(\ell, j, k) \in E_c$. The *unipotent subgroup* is $U = \langle T_U \rangle$.

Theorem 1.5. *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ be arbitrary. For every $g \in G_c(\mathfrak{m})$ there exist $h \in H$ and $u \in U$ with $g = hu$.*

The inclusion $HU \subseteq G_c(\mathfrak{m})$ is trivial, so the theorem says exactly that $G_c(\mathfrak{m}) = HU$. The source states this for c given by the coefficients of J . The formal statement holds for every c , in particular for that one, and needs no hypothesis such as $c(2) \geq 1$.

2 Lemmas

2.1 The torus normalizes the unipotent subgroup

The listed lemmas rely on Theorem 3.4 and Lemma 3.5 of the source. Both are machine-checked in the same development, and we give their proofs here.

Lemma 2.1 (H normalizes U ; Theorem 3.4 of the source). *For every c , every $h \in H$ and every $u \in U$, we have $huh^{-1} \in U$.*

Proof. Step 1: conjugating generators. For $t \in T_H$ and $g \in T_U$ we have $tgt^{-1} \in T_U$. This is a case check using the defining relations. For $t = H_1(s)$:

$$\begin{aligned} H_1(s)X_{-1}(u)H_1(s)^{-1} &= X_{-1}(su) && \text{by (Re:6c's companion (Re:6a))}, \\ H_1(s)Y_{-1}(u)H_1(s)^{-1} &= Y_{-1}(s^{-1}u) && \text{by (Re:6c)}, \\ H_1(s)X_{\ell,jk}(u)H_1(s)^{-1} &= X_{\ell,jk}(s^{\ell+1}u) && \text{by (Im:6a)}, \\ H_1(s)Y_{\ell,jk}(u)H_1(s)^{-1} &= Y_{\ell,jk}((s^{\ell+1})^{-1}u) && \text{by (Im:6c)}. \end{aligned}$$

For $t = H_2(s)$ one uses (Re:6b), (Re:6d), (Im:6b) and (Im:6d) in the same way. In each case the result is again a unipotent generator.

Step 2: from generators to U . Fix $t \in T_H$ and let $V = \{u \in U \mid tut^{-1} \in U\}$. By Step 1, $T_U \subseteq V$, and $1 \in V$. The set V is closed under products and inverses, because

$$t(ab)t^{-1} = (tat^{-1})(tbt^{-1}), \quad ta^{-1}t^{-1} = (tat^{-1})^{-1}.$$

Hence $V = U$, that is, $tUt^{-1} \subseteq U$.

Step 3: T_H is closed under inverses.

- By (H:1a) with $s = t = 1$ we get $H_1(1)^2 = H_1(1)$, so $H_1(1) = 1$.
- By (H:1a) again, $H_1(s)H_1(s^{-1}) = H_1(1) = 1$, so $H_1(s^{-1}) = H_1(s)^{-1}$.

- By (Re:5a), $H_2(s) = \tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1}$. Therefore

$$H_2(s^{-1}) = \tilde{w}_{-1}H_1(s)^{-1}\tilde{w}_{-1}^{-1} = H_2(s)^{-1}.$$

Step 4: from T_H to H . By Step 3, every element of H is either 1 or a product ty with $t \in T_H$ and $y \in H$ a shorter such product. We argue by induction on the length of the product. The claim is trivial for 1. If $yUy^{-1} \subseteq U$, then for $u \in U$,

$$(ty)u(ty)^{-1} = t(yuy^{-1})t^{-1} \in U$$

by Step 2. This proves $hUh^{-1} \subseteq U$ for all $h \in H$. $\square$

Lemma 2.2 (Moving U past H ; Lemma 3.5 of the source). *For every c , every $h \in H$ and every $u \in U$, there exists $u' \in U$ with $uh = hu'$.*

Proof. Take $u' = h^{-1}uh = h^{-1}u(h^{-1})^{-1}$. Since $h^{-1} \in H$, the preceding lemma (Theorem 3.4 of the source) gives $u' \in U$. Clearly $hu' = uh$. $\square$

2.2 The set HU is a subgroup

Lemma 2.3 (Closure under products). *Let $a, b \in G_c(\mathfrak{m})$. Suppose $a = h_1u_1$ and $b = h_2u_2$ with $h_1, h_2 \in H$ and $u_1, u_2 \in U$. Then there exist $h \in H$ and $u \in U$ with $ab = hu$.*

Proof. By Lemma 3.5 of the source (proved above) applied to h_2 and u_1 , there is $u'_1 \in U$ with $u_1h_2 = h_2u'_1$. Then

$$ab = h_1(u_1h_2)u_2 = h_1(h_2u'_1)u_2 = (h_1h_2)(u'_1u_2).$$

Here $h_1h_2 \in H$ and $u'_1u_2 \in U$. $\square$

Lemma 2.4 (Closure under inverses). *Let $a \in G_c(\mathfrak{m})$ with $a = hu$, where $h \in H$ and $u \in U$. Then there exist $h' \in H$ and $u' \in U$ with $a^{-1} = h'u'$.*

Proof. We have $h^{-1} \in H$ and $u^{-1} \in U$. By Lemma 3.5 of the source (proved above) applied to h^{-1} and u^{-1} , there is $u' \in U$ with $u^{-1}h^{-1} = h^{-1}u'$. Hence

$$a^{-1} = u^{-1}h^{-1} = h^{-1}u'.$$

Take $h' = h^{-1}$. $\square$

Lemma 2.5 (Identity). *There exist $h \in H$ and $u \in U$ with $1 = hu$ in $G_c(\mathfrak{m})$.*

Proof. Take $h = 1 \in H$ and $u = 1 \in U$. $\square$

2.3 Elements of H , of U , and generators lie in HU

Lemma 2.6 (Toral elements). *If $x \in H$, then there exist $h \in H$ and $u \in U$ with $x = hu$.*

Proof. Take $h = x$ and $u = 1$. $\square$

Lemma 2.7 (Unipotent elements). *If $x \in U$, then there exist $h \in H$ and $u \in U$ with $x = hu$.*

Proof. Take $h = 1$ and $u = x$. $\square$

Lemma 2.8 (Generators). *For every symbol $g \in \mathcal{X}$, the image of g in $G_c(\mathfrak{m})$ can be written as hu with $h \in H$ and $u \in U$.*

Proof. We distinguish the type of the symbol.

- If $g = H_1(s)$ or $g = H_2(s)$, its image lies in $T_H \subseteq H$. Write it as $g \cdot 1$.
- If g is one of $X_{-1}(u)$, $Y_{-1}(u)$, $X_{\ell,jk}(u)$, $Y_{\ell,jk}(u)$, its image lies in $T_U \subseteq U$. Write it as $1 \cdot g$. $\square$

3 Proof of the theorem

Proof of Theorem 1.5. Let $S = \{x \in G_c(\mathfrak{m}) \mid x = hu \text{ for some } h \in H, u \in U\}$. Then:

- S contains 1 by Lemma 2.5;
- S is closed under products by Lemma 2.3;
- S is closed under inverses by Lemma 2.4.

Hence S is a subgroup of $G_c(\mathfrak{m})$. By Lemma 2.8, S contains the image of every generator in $\mathcal{X}$. The group $G_c(\mathfrak{m}) = F(\mathcal{X})/N_{\mathcal{R}}$ is generated by these images, since every element is the image of a word in the generators. Hence the subgroup S is all of $G_c(\mathfrak{m})$, and every $g \in G_c(\mathfrak{m})$ has the form $g = hu$ with $h \in H$ and $u \in U$. $\square$

Relation to the source. The argument is the source proof: show that HU is a subgroup and that it contains all generators. The formalization deviates only in the following points.

- It also checks that $1 \in HU$ (Lemma 2.5).
- It makes explicit that $G_c(\mathfrak{m})$ is generated by the images of its generators.
- It works with an arbitrary $c: \mathbb{N} \rightarrow \mathbb{N}$ instead of the coefficients of J .
- It proves Lemma 3.5 of the source from Theorem 3.4 of the source (both stated and proved above). Theorem 3.4 is in turn established from the relations (Re:6a)–(Re:6d), (Im:6a)–(Im:6d), (H:1a) and (Re:5a).

Lemmas 2.6 and 2.7 are verified but not needed in the final argument.

A Correspondence with the formal proof

Each lemma of the text and the Lean declaration that states it; Theorem 1.5 is `gsimple_g_m_eq_h_u`.

Lemma	Lean declaration	Status
Lemma 2.3	<code>hu_mul</code>	verified
Lemma 2.4	<code>hu_inv</code>	verified
Lemma 2.5	<code>hu_one</code>	verified
Lemma 2.6	<code>hu_of_torus</code>	verified
Lemma 2.7	<code>hu_of_unip</code>	verified
Lemma 2.8	<code>hu_gen</code>	verified